OXFORD
INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL A-LEVEL Biology

(9610)

Unit 2: Biological systems and disease

Mark Scheme

June 2017

Version: 1.0 Final

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Copyright © 2017 Oxford International AQA Examinations and its licensors. All rights reserved.

Question	Marking Guidance	Mark	Comments
01.1	(Push down firmly) to produce a thin layer (of tissue) / so that light can pass through; (coverslip does not move sideways) to prevent damage to chromosomes/chromatids;	2	
01.2	Any **two** from: Clean the objective lens; Adjust the (coarse/fine) focus; Select a different objective lens; Adjust the condenser;	2	 Allow increase the magnification Allow let more light through
01.3	(D) C E B A	1	
01.4	B; Chromatids are being pulled apart	2	 Allow to opposite ends/poles for 'apart'
01.5	0.4625;;	2	Allow 46% Accept 0.46/0.463 Reject 0.462 Allow 1 mark for $\frac{(21 + 8 + 5 + 3)}{80}$ or $\frac{37}{80}$ but no answer / incorrect answer Allow answer as ratio (0.86:1)

01.6	72;;	2	Allow 1 mark for 8/80 × 720 but no answer / incorrect answer Also allow 1 mark for $\frac{8 \times 12 \times 60}{80}$ but no answer / incorrect answer

Question	Marking guidance	Mark	Comments
02.1	Fatty acids and monoglycerides move in by (simple/passive) diffusion; Description of moving out via vesicle fusing with cell membrane;	2	Reject facilitated diffusion 'Bulk transport' is insufficient 'In vesicles' is insufficient Allow exocytosis
02.2	Emulsification;	1	
02.3	Triglycerides converted to fatty acids; Fatty acids decrease pH / increase acidity; (Increased acidity) causes indicator to change from pink to colourless;	3	
02.4	Tubes **1** and **2**: To show that lipase is needed for digestion of triglycerides **o**r to show that bile salts do not digest triglycerides; Tubes **3** and **4**: To show that lipase is an enzyme **or** lipase structure is destroyed by boiling; Tubes **3** and **5**: To show that bile salt is not an enzyme / a protein;	3	Allow to see the effect of lipase in breaking down triglycerides Allow lipase is denatured by boiling Allow to see if lipase is an enzyme Allow bile salts are not affected by boiling
02.5	Tube **3** (with bile salts) reacted faster than tube **1** (without bile salts);	1	Allow correct reference to tube **5** instead of tube **3** Allow lipase alone took 8 mins but lipase + bile only took 6 mins

Question	Marking guidance	Mark	Comments
03.1	Aorta has thicker wall (than the vena cava);	2 max	Answer must be comparative Allow aorta only has a thick wall compared to the vena cava.
	Aorta has a narrower lumen (than the vena cava);		Answer must be comparative Allow aorta only has a narrow lumen.
	Aorta contains more (smooth) muscle (than the vena cava);		Answer must be comparative Accept aorta contains lots of (smooth) muscle
	Aorta contains more elastic tissue (than the vena cava);		Answer must be comparative Accept aorta contains lots of elastic tissue Reject aorta does not have valves unless clearly along the length (except for semi-lunar) Ignore references to oxygenated/deoxygenated blood and blood pressure Allow converse answers for vena cava

03.2	Narrower vessels; (Provides) more friction/resistance (to blood flow); OR Larger total cross sectional area (of capillaries); Same volume of blood per unit time / same volume flow rate spread out over larger (total) area;	2	 Answer must be comparative Answer must be comparative

03.3	(More) time for exchange of substances	1	

03.4	Leg muscles contract; Increases pressure of blood (in vein); Forcing/pushing blood up(wards) through upper/top valve; Forcing/pushing blood back onto lower/bottom valve which closes it (preventing blood moving back down);	4	Reject muscles constrict Allow blood pressure goes up Answer must refer to blood being forced/pushed through valve Answer must refer to blood being forced/pushed onto lower valve

Question	Marking guidance	Mark	Comments
04.1	Interphase;	1	
04.2	Increase in protein content; Organelles increase in number/size/replicate	2	Accept (increase in) protein synthesis. Allow mitochondria or ribosomes as named organelles here
04.3	S phase;	1	Accept synthesis phase
04.4	6(%);	1	
04.5	1.84;;	2	Correct answer = 2 marks Allow 1 mark for use of 0.94 and 0.51 but wrong answer / no answer Allow 1 mark if too many significant figures Allow 1 mark for answer correct for student's figures from the graph
04.6	(They/cancer cells take a) longer time to recover; (They/cancer cells have a) reduced rate of mitosis / divide more slowly; Higher proportion of cancer cells (than healthy cells) affected by cytarabine as cancer cells divide more frequently;	2 max	Answer must be comparative Answer must indicate a change in rate Accept (they/cancer cells) have an increased doubling time Accept: (they/cancer cells) unable to synthesise DNA Allow other sensible suggestions – eg cancer cells absorb more cytarabine than healthy cells

Question	Marking Guidance	Mark	Comments
05.1	Less resistance to flow / easier flow;	1	
05.2	Less/no ATP produced / Less/no energy released; Less/no organic substances transported into sieve cell / Less/no organic substances transported out of companion cell; By <u>active transport</u>;	3	Reject less/no energy produced Accept sucrose or amino acid as named organic substance
05.3	1. Fluid is under pressure; 2. (Sucrose) can be transported both up and down the plant stem;	2	Allow in both directions for 'up and down'
05.4	Distance between X and Y; Time (for radioactivity) to travel between X and Y;	2	If distance **and** time both referred to with no mention of 'between X and Y' allow 1 mark.
05.5	D;	1	

Question	Marking guidance	Mark	Comments
06.1	To allow comparison;	1	Allow to see how heart rate increases from rest Allow to see if there is a change/difference in heart rate
06.2	Idea of using a fixed duration of exercise; Idea of using a fixed/standard number of repeats; Description of an action taken to **standardise exercise**;	2 max	Accept idea of exercising for a set period of time. Accept idea of having a fixed rate of repeating. Allow push-ups from a fixed position; complete squats from a standing start. Allow same intensity of exercise. Ignore 'say what exercise'
06.3	Idea of not measuring complete beats; Small difference in measurements can produce large difference in pulse rate;	1 max	Allow idea of not measuring complete cycles Accept idea that any measurement/counting errors are exaggerated when multiplying up
06.4	120 (cm^3 $beat^{-1}$);	1	
06.5	284(%);	1	

06.6	Supplies <u>more</u> oxygen/glucose (to the muscles); **SO** <u>More</u> respiration / <u>more</u> energy released / <u>more</u> ATP produced; Removes <u>more</u> carbon dioxide / lactate; **SO** Acidity does not increase (too much) / pH does not fall (too much); <u>More</u> heat can be carried away (from the muscle);	3 max	Must be comparative Allow supplies oxygen/glucose quicker Must be comparative Reject (more) energy made/created Must be comparative Allow removes carbon dioxide/lactate quicker Allow less lactic acid formed Allow reduces formation of lactic acid Allow carbon dioxide/lactate increase acidity / decrease pH Must be comparative Allow heat can be carried away faster

Question	Marking guidance	Mark	Comments
07.1	(Protein on the surface of the Ebola virus that) Causes/generates an immune response **OR** Causes/generates antibody production;	1	Ignore 'protein on the surface' as in stem Allow stimulates the immune system Allow recognised as non-self by immune system
07.2	Agglutination;	1	Accept phonetically correct spelling (Not aggulation)
07.3	Allows phagocytosis to be more efficient/work better **OR** Precipitates out soluble toxins;	1	Accept stimulates phagocytosis/inflammation
07.4	Not all of the (injected) antibodies are specific to (the) Ebola (antigen/virus); The (injected) antibodies are seen as foreign antigens and the body produces an immune response against them; The (injected) antibodies only last for a short time in the blood;	2 max	Allow polyclonal rather than monoclonal antibodies were injected Accept body produces antibody against the Ebola antibody. Reject any reference to antibodies being rejected. Accept there are no memory cells to keep replacing the injected antibodies.

07.5	1 B-cells / plasma cells make specific antibodies; 2 Retention of ability to make specific antibody; 3 Reduced number of susceptible hosts;	3	Allow ‘active immunity’ Allow ‘lymphocytes’ for B-cells / plasma cells’ Allow ‘immunological memory’. Allow idea of memory cells. Allow ‘secondary response’. Allow idea of ‘herd immunity’

Question	Marking Guidance	Mark	Comments
08.1	Water (from soil) enters root hair (cell)/epidermis of root/root epidermal cells; (Water) travels through the cortex; By the symplastic pathway/through the cytoplasm; By the apoplastic pathway/through the cell walls;	4	
08.2	(Increased air temperature increases rate because) kinetic energy (of water molecules) increases; (so) faster movement/diffusion (out of the leaf); (Increased light intensity increases rate because) increased number of stomata open/increased size of stomata; (so) larger S.A. for water (vapour) movement/diffusion; (Increasing humidity decreases rate because) Decreased water potential gradient/diffusion gradient (between the air spaces in the leaf and the atmosphere); Slower diffusion;	6	Mark as pairs Alternative to 'increased air temperature' – warm air can hold more water vapour so more water can evaporate off the leaf surfaces